MATH329 · OPERATIONS RESEARCH

Big-M Practice

6 problems: from the slides, quiz 471, and practice

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01 Questions

For each problem: (a) write the standard form and add the artificial variables. (b) Solve it with the Big-M method. The full solution is in a separate file.

1 MIN · SLIDES, EXAMPLE 1

Minimize $z = 2x_1 + x_2$

Subject to $3x_1 + x_2 \geq 9$

$x_1 + x_2 \geq 6$

$x_1, x_2 \geq 0$

2 MAX · SLIDES, EXAMPLE 2

Maximize $z = 2x_1 + 3x_2 - 5x_3$

Subject to $x_1 + x_2 + x_3 = 7$

$2x_1 - 5x_2 + x_3 \geq 10$

$x_1, x_2, x_3 \geq 0$

3 MAX · QUIZ 3-4 (471)

Maximize $z = 2x_1 + 3x_2 - 5x_3$

Subject to $4x_1 + 3x_2 - x_3 \leq 24$

$x_1 + x_2 + 2x_3 \geq 18$

$x_1, x_2, x_3 \geq 0$

4 MIN · = , ≥ , ≤

Minimize $z = 4x_1 + x_2$

Subject to $3x_1 + x_2 = 3$

$4x_1 + 3x_2 \geq 6$

$x_1 + 2x_2 \leq 4$

$x_1, x_2 \geq 0$

5 MAX · ≤ , ≥

Maximize $z = 3x_1 + 2x_2$

Subject to $x_1 + x_2 \leq 4$

$x_1 + 3x_2 \geq 6$

$x_1, x_2 \geq 0$

6 MAX · SPECIAL CASE

Maximize $z = 2x_1 + x_2$

Subject to $x_1 + x_2 \leq 2$

$x_1 + x_2 \geq 4$

$x_1, x_2 \geq 0$

What does the final tableau tell you about this problem?